

Test_2



1. Conditions (if, elif, else)

1. Write a program to check whether a number is positive, negative, or zero.

```
num = int(input("Enter Number : "))

if num>0:
    print(f"{num} is positive")
elif num<0:
    print(f"{num} is Negative")
else:
    print(f"{num} is zero")
```



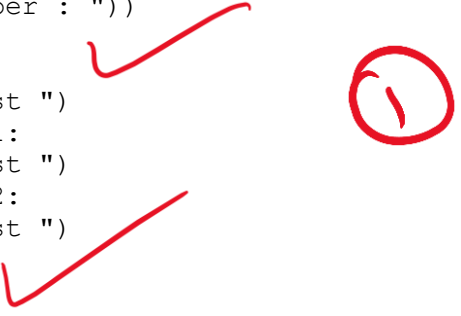
Enter Number : 3

3 is positive

2. Write a program to find the largest of three numbers.

```
num1 = int(input("Enter Number : "))
num2 = int(input("Enter Number : "))
num3 = int(input("Enter Number : "))

if num1>num2 and num1>num3:
    print(f"{num1} is Largest ")
elif num2>num3 and num2>num1:
    print(f"{num2} is Largest ")
elif num3>num1 and num3>num2:
    print(f"{num3} is Largest ")
else:
    print("All are Equal ")
```



Enter Number : 12
Enter Number : 12
Enter Number : 12

All are Equal

3. Write a program to check whether a given year is a leap year.

```
year = int(input("Enter the Year : "))

if year%400 == 0:
    print(f"{year} is leap year")
elif year%100 == 0:
    print(f"{year} is not leap year")
elif year % 4==0:
    print(f"{year} is leap year")
else:
    print(f"{year} is not leap year")
```



Enter the Year : 2000

2000 is leap year

4. Write a program to check if a character is a vowel or consonant.

```
ch = input("Enter a character : ")
vow="aeiouAEIOU"
```

```
if ch in vow:
    print(f"{ch} is vowel ")
else:
    print(f"{ch} is consonant")
```

Enter a character : w

w is consonant

5. Write a program to calculate the electricity bill:

- First 100 units → ₹5/unit
- Next 100 units → ₹7/unit
- Above 200 units → ₹10/unit

```
unit = int(input("Enter a consumed unit : "))
```

```
if unit<=100:
    bill = unit * 5
    print("Bill Amount is ",bill)
elif unit>100 and unit<=200:
    bill = unit * 7
    print("Bill Amount is ",bill)
else:
    bill = unit * 10
    print("Bill Amount is ",bill)
```

Enter a consumed unit : 210

Bill Amount is 2100

2. Loops (for / while)

1. Print all even numbers from 1 to 100.

```
for i in range(1,101):
    if i%2==0:
        print(i,end = " ")
```

2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48
50 52 54 56 58 60 62 64 66 68 70 72 74 76 78 80 82 84 86 88 90 92 94 96
98 100

2. Find the factorial of a given number using a loop.

```
num = int(input("Enter a Number : "))
```

```
fact = 1
for i in range(1,num+1):
    fact = fact*i
```

```
print("Factorial is ",fact)
```

```
Enter a Number : 5
```

```
Factorial is 120
```

3. Print the Fibonacci series up to n terms.

```
num = int(input("Enter a Number : "))
```

```
a=0
```

```
b=1
```

```
for i in range(1,num+1):  
    print(a,end=" ")  
    a,b=b,a+b
```

```
Enter a Number : 7
```

```
0 1 1 2 3 5 8
```

4. Find the sum of digits of a given number.

```
num = int(input("Enter a Number : "))
```

```
temp = num
```

```
sum1 = 0
```

```
while temp>0:
```

```
    rem = temp%10
```

```
    sum1 = sum1 + rem
```

```
    temp =temp//10
```

```
print(sum1)
```

```
Enter a Number : 1234
```

```
10
```

5. Check whether a number is an Armstrong number

```
num = int(input("Enter a Number :"))
```

```
digit =len(str(num))
```

```
temp = num
```

```
sum1 = 0
```

```
while temp>0:
```

```
    digits = temp%10
```

```
    sum1 = sum1+ digits**digit
```

```
    temp =temp//10
```

```
if num == sum1:
```

```
    print("Armstrong")
```

```
else:
```

```
    print("Not Armstrong")
```

```
Enter a Number : 123
```

```
Not Armstrong
```

3. Functions

1. Write a function to find the greatest of three numbers.

```
def greatest(num1,num2,num3):
    if num1>num2 and num1>num3:
        print(f"{num1} is Largest ")
    elif num2>num3 and num2>num1:
        print(f"{num2} is Largest ")
    elif num3>num1 and num3>num2:
        print(f"{num3} is Largest ")
    else:
        print("All are Equal ")
```

greatest(30,40,20)

40 is Largest

2. Write a function to check whether a number is prime.

```
def prime(n):
    is_Prime=True

    if n<=1:
        print(f"{n} is not Prime")
    for i in range(2,n):
        if n%i == 0:
            is_Prime = False

    if is_Prime:
        print(f"{n} is Prime")
    else:
        print(f"{n} is not Prime")
```

n = int(input("Enter a Number :"))
prime(n)

Enter a Number : 6

6 is not Prime

3. Write a function that returns the reverse of a string.

```
def rev(str1):
    st=""
    for i in str1:
        st = i+st
    print(st)
```

str1 = input("Enter a Number :")
rev(str1)

Enter a Number : wer

rew

4. Write a function to calculate the area and perimeter of a rectangle.

```
def area_peri(l,b):
    area = l*b
    peri=2*(l+b)

    print("Area is ",area)
    print("Perimeter is ",peri)

l= int(input("Enter Length : "))
b = int(input("Enter Breadth : "))

area_peri(l,b)

Enter Length : 3
Enter Breadth : 4

Area is 12
Perimeter is 14
```



5. Write a function that accepts a list and returns the second largest element.

```
def s_lar(lst):
    maxi = lst[0]
    sl=lst[0]

    for i in lst:
        if i>maxi:
            sl = maxi
            maxi = i
        elif i>sl and i!=maxi:
            sl = i
    print("Second Largest is ",sl)
```



```
lst = [1,3,4,5,7,8,34,98]
s_lar(lst)

Second Largest is 34
```

4. Exception Handling

1. Write a program to handle ZeroDivisionError.

```
try:
    a = int(input("Enter a Number 1"))
    b = int(input("Enter a Number 2"))

    print(a/b)
except ZeroDivisionError:
    print("Cannot Divide by Zero ")
except ValueError:
    print("Enter Valid Number")
```



```
Enter a Number 1 23
Enter a Number 2 0
```

Cannot Divide by Zero

2. Write a program to handle ValueError when the user enters invalid input.

```
try:
    a = int(input("Enter a Number 1: "))

    print(a)
except ValueError:
    print("Enter Valid Number")
```

Enter a Number 1: s

Enter Valid Number

3. Create a custom exception to check if age is less than 18.

```
class AgeError (Exception):
    pass
```

```
try:
    age = int(input("Enter Your Age : "))
    if age<18:
        raise AgeError("Invalid Age, Please Enter age above 18 ")
```

```
except AgeError as e:
    print(e)
else:
    print("Eligible")
finally:
    print("Thank You")
```

Enter Your Age : 12

Invalid Age, Please Enter age above 18
Thank You

4. Write a program that handles both IndexError and KeyError.

```
try:
    lst = [10, 20, 30]
    print(lst[5])

except IndexError:
    print("Index Out Of Range")
```

```
try:
    d = {"name": "Rahul"}
    print(d["age"])
```

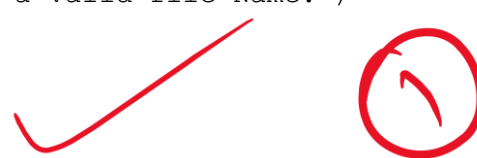
```
except KeyError:
    print("Key Not Found")
```

Index Out Of Range
Key Not Found

(1)
(IndexError, KeyError)

5. Write a program using try, except, else, and finally to read a file and display its contents.

```
try:
    file = open("D:\\Python_Test_Questions-1.txt",'r')
except FileNotFoundError:
    print("File Name Error! Please Enter a valid file Name.")
else:
    print("Output is:\n",file.read())
finally:
    print("Thank You")
```



Output is:
Python Test Questions

1. Conditions (if, elif, else)

1. Write a program to check whether a number is positive, negative, or zero.
2. Write a program to find the largest of three numbers.
3. Write a program to check whether a given year is a leap year.
4. Write a program to check if a character is a vowel or consonant.
5. Write a program to calculate the electricity bill:
- First 100 units @' 15/unit
 - Next 100 units @' 17/unit
 - Above 200 units @' 10/unit

2. Loops (for / while)

1. Print all even numbers from 1 to 100.
2. Find the factorial of a given number using a loop.
3. Print the Fibonacci series up to n terms.
4. Find the sum of digits of a given number.
5. Check whether a number is an Armstrong number.

3. Functions

1. Write a function to find the greatest of three numbers.
2. Write a function to check whether a number is prime.
3. Write a function that returns the reverse of a string.
4. Write a function to calculate the area and perimeter of a rectangle.
5. Write a function that accepts a list and returns the second largest element.

4. Exception Handling

1. Write a program to handle ZeroDivisionError.
2. Write a program to handle ValueError when the user enters invalid input.
3. Create a custom exception to check if age is less than 18.
4. Write a program that handles both IndexError and KeyError.
5. Write a program using try, except, else, and finally to read a file and display its contents.

Code Executed Successfully